

Predict2Act: An AI-Driven Sleep Irregularity Prediction and Intervention System for College Students Using Oura Ring Data

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Project Summary

This project will develop a personalized intelligent system that uses recent historical Oura Ring data to predict a college student's next-day sleep irregularity before it occurs and then select one specific, measurable behavioral intervention. Rather than calculating another current sleep metric or generating broad sleep advice, the system adds two key capabilities: (1) future-state prediction from historical data and (2) actionable intervention selection guided by an evidence-based sleep framework.

Past 7-day Oura data → ML predicts next-day sleep irregularity → CBT-I-informed actionable intervention → Next-day Oura outcome

1. Motivation

Sleep irregularity is a meaningful problem among college students because inconsistent sleep–wake schedules are associated with poorer academic performance and delayed circadian timing [1]. College students are especially vulnerable to irregular sleep due to changing class schedules, academic workload, and late-night activities. Yet sleep problems are often recognized only after they occur, and general sleep advice may not translate into a clear action for the individual. This creates a practical need to move from retrospective sleep tracking toward proactive prediction and specific, actionable intervention.

2. Goal

Use at least three months of daily Oura history to predict next-day sleep irregularity. For each upcoming night the model predicts three quantities — bedtime, wake time, and sleep duration, each as a signed deviation in minutes from the student's own 28-day rolling baseline. The Sleep Irregularity Score (SIS) is a frozen, deterministic function of those three, defined before any evaluation. Predicting components rather than a single composite keeps the output in interpretable units and allows the largest predicted deviation to identify which behavior to target. SIS is distinct from the Sleep Regularity Index, which compares sleep–wake state 24 hours apart across a window, from –100 to 100[1]. The core research question is: can the previous seven days of Oura data predict a college student's next-day sleep irregularity?

3. Existing Systems

Existing work provides strong foundations in sleep sensing, sleep prediction, sleep regularity measurement, and behavioral intervention. Just-in-time adaptive interventions (JTAs) demonstrate that personalized and timely sleep support is feasible [3], though prior work notes challenges in specifying receptivity and decision rules [2]. Wearable machine learning systems are another important system category. Studies show that future health-related states can be forecast from physiological and behavioral data using personalized machine learning [4]. However, forecasting sleep irregularity is a relatively understudied domain [5]. Another important category is irregularity quantification frameworks, which can capture day-to-day sleep regularity [1]. However, there is not yet a research consensus on what threshold defines sleep irregularity [7]. The final category is college student sleep monitoring studies, which connect irregular sleep with academic and well-being outcomes [1], though observational findings and associations can vary based on the sleep metric of focus [6]. Overall, limited prior work was found focusing on predicting personalized next-day sleep irregularity and using those predictions to drive a specific and measurable intervention.

4. Proposed System

Historical Oura data will be exported in structured form via the Oura API. For each day, a rolling seven-day window is expressed as deviations from the student's personal 28-day baseline and includes recent sleep timing and duration, their short-term variability, cumulative sleep debt, the day of week of the target night, and Oura physiological and activity signals such as heart rate variability, resting heart rate, and daily activity. We will train regularized linear regression (Lasso), which performs feature selection and yields interpretable coefficients, and compare it against gradient-boosted trees; with roughly 100 nights of personal history this is a small-data regime, so deep architectures are not appropriate. Models are validated using rolling-origin (expanding-window) splits rather than random cross-validation, and all baselines and preprocessing are fit on training data only, because the prediction is issued before the upcoming sleep period. The largest predicted component deviation then selects one rule from a constrained, CBT-I-informed intervention set [8]. For example, instead of “keep a regular schedule,” the system recommends “keep tomorrow's wake time within ± 30 minutes of your usual time.” The following day's Oura data supplies the observed outcome. The same pipeline will also be run on a public multi-person wearable dataset [9] to test whether the method generalizes beyond a single device and cohort.

5. Benefits / Key Contribution

The system is predictive, personalized, action-oriented, and directly evaluable. Its main contribution is the combination of historical-data forecasting with a structured intervention step: Historical Oura Data → Predict Future Sleep Irregularity → Deliver One Actionable Intervention → Measure Outcome. Success is defined concretely: the model must reduce mean absolute error on held-out, chronologically later wake-time deviation relative to three baselines — the personal mean, the previous day's value, and the personal day-of-week mean — with the day-of-week baseline as the meaningful bar, since weekday and weekend patterns explain much of the variance in sleep timing. Intervention efficacy is not claimed; establishing it would require a controlled comparison and is left to future work. The project is intended for sleep wellness and decision support, not for diagnosing or treating insomnia.

References

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